

Keenan Elliott, MPH, MSc

Epidemiologist | Research Scientist

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SUMMARY

- Epidemiologist with 3+ years of experience coordinating and executing public health research
- Experienced at applying statistical modelling & epidemiological methods for population-based analyses
- Skilled in Python, R, & SQL for data extraction, cleaning, & analysis to support evidence-based decision-making
- Proven ability to develop automated data pipelines & reproducible workflows that inform decision making
- Experienced at collaborating with academic, government, and health authority partners for public health research

TECHNICAL SKILLS

Languages & Tools : R, Python, SQL, Git/GitHub, Excel, LaTeX, Bash, Unix

Experimental Design : Time-Series Analysis, Simulation Studies, Quasi-experimental studies, Hypothesis Testing

Statistical & Epidemiological Methods : Generalized Linear Models, Mixed Effects Models, Survival Analysis, Causal Inference, Imputation/Missing Data Methods, Growth Mixture Models, Principal Component Analysis

EXPERIENCE

Stringhini Lab, University of British Columbia

05/2025 – Present

Research Scientist

Vancouver, BC

- Lead analyses of longitudinal cohorts to identify drivers of chronic disease and health inequities
- Apply causal inference methods to assess biological & environmental determinants of health
- Maintain databases & develop strategies to handle missing data in multi-year cohort studies
- Support lab members & graduate students with analysis development and reproducible workflows using Git
- Produce high-impact articles to inform public policy aimed at reducing population health inequities

BC Centre for Disease Control

09/2024 – 12/2024

Research Analyst (Internship)

Vancouver, BC

- Screened & appraised >1500 articles for systematic review on upstream determinants of alcohol use
- Systematized procedures for data extraction, quality appraisal, & synthesis of evidence
- Collaborated with internal & external stakeholders to support evidence-based interventions & policies

Public Health Agency of Canada

05/2024 – 09/2024

Data Analyst (Internship)

Vancouver, BC

- Developed & maintained national wastewater systems to enhance infectious disease surveillance
- Automated data extraction, cleaning, & analysis tools to improve outbreak detection & surveillance
- Conducted statistical and descriptive analyses on time-series data to identify emerging disease trends
- Developed knowledge translation products & dashboards for stakeholder communication & planning

Lowenberger Lab, Simon Fraser University

09/2020 – 06/2023

Research Assistant

Burnaby, BC

- Conducted molecular & computational experiments to uncover determinants of disease transmission
- Used bioinformatics & machine learning methods in Bash, R, & Python to analyze multi-omics data
- Authored manuscripts & developed data visualizations for publication in high-impact journals

SELECTED PROJECTS

Health Data Integration & Quality Assurance (Longitudinal Ageing Studies) – UBC

- Lead an international longitudinal data harmonization project involving ~500,000 participants across five nations
- Standardize & validate data integrity across cohorts to ensure dataset integration
- Coordinate data access applications and data-sharing agreements with international research partners
- Manage the complete data pipeline, including cleaning, reproducible analyses, & reporting
- Contributed to securing and managing a successful multi-year CIHR grant

Canadian Wastewater Modelling Report – Public Health Agency of Canada

- Drastically reduced time to produce pathogen surveillance reports, from hours to minutes, through an end-to-end automated scripting system for the National Wastewater Monitoring Program
- Used SQL & R in an Azure environment to produce daily, weekly, & monthly detailed reports, guiding public health infrastructure planning

Understanding the Immune & Metabolic Drivers of Host-Virus Compatibility – SFU

- Led a multi-year project, from conceptualization to analysis, validation, visualization & reporting
- Executed comprehensive bioinformatic & machine learning analyses
- Synthesized complex metabolomics & immune phenotyping results into primary academic work published in high-impact, peer-reviewed journals

EDUCATION

University of Toronto

09/2023 – 12/2024

MPH, Epidemiology

Toronto, ON

- **Relevant Coursework:** Communicable Disease Epidemiology, Epidemiologic Methods, Biostatistics, Modern Statistical Genetics, Analysis of Correlated Data

Simon Fraser University

09/2020 – 12/2022

MSc, Biological Sciences (Parasitology)

Burnaby, BC

- **Thesis:** Insights into host-virus interactions: Using a metabolomics approach to study *Aedes aegypti*-dengue virus compatibility

Simon Fraser University

09/2013 – 05/2018

BSc, Molecular Biology & Biochemistry

Burnaby, BC

GPA: 3.37

PUBLICATIONS

- **Elliott K.**, Caicedo PA, Haunerland NH, Lowenberger C, (2023). Profiling lipidomic changes in dengue-resistant and dengue-susceptible strains of Colombian *Aedes aegypti* after dengue virus challenge. *PLoS Negl Trop Dis* 17(10):e0011676
- Marijan, D., Tse, R., **Elliott, K.**, Chandhok, S., Luo, M., Lacroix, E., Audas, T. (2019). Stress-specific aggregation of proteins in the amyloid bodies. *FEBS Letters*, 593(22): 3163-3172
- Hartwell, R., Chan, B., **Elliott, K.**, Alnojeidi, H., Ghahary, A. (2016). Polyvinyl alcohol-graft-polyethylene glycol hydrogels improve utility and biofunctionality of injectable collagen biomaterials. *Biomedical Materials*, 11(3)
- Vereecke S, **Elliott K**, Schrepft S, Jarvis I, Gemmell E, Brauer M, Kobor M, Murphy R, Valeri L, Stringhini S (Under Review). Social and Environmental Inequalities in Functional Aging among Older Adults: A Mediation Analysis from the Canadian Longitudinal Study on Aging (CLSA)
- **Elliott K.**, Caicedo PA, Haunerland NH, Lowenberger C, (In progress). Metabolomic profiling of dengue virus-resistant and susceptible *Aedes aegypti*. *Current Research in Parasitology and Vector-Borne Diseases*